Hands-On Activity 10 | Explaining Visually to Influence with Data

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Course Code and Title : CPE-031 - Visualizations and Data Analysis
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1. Objectives:

This activity aims to help students demonstrate how to influence and communicate effectively with data by identifying audience needs, crafting a clear message, and telling a story visually through charts.

2. Intended Learning Outcomes (ILOs):

By the end of this activity, students should be able to:

1. Analyze the audience to understand their level of data knowledge and needs.
2. Evaluate how well a message and visualization communicate insights.
3. Create a clear and meaningful data story using Python visualization tools.

3. Discussions:

Data storytelling goes beyond showing charts it's about influencing decisions and sparking understanding.

Know Your Audience: Decision-makers may prefer clear summaries, while analysts like detailed data.

Know Your Message: Focus on the why it matters part of the story.

Tell a Story: Use preattentive attributes color, size, and position to guide attention.

Structure Your Story: Like Aristotle's rhetorical triangle, combine logic (logos), emotion (pathos), and credibility (ethos).

4. Procedures:

Step 1. Import Libraries and Create Sample Data

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

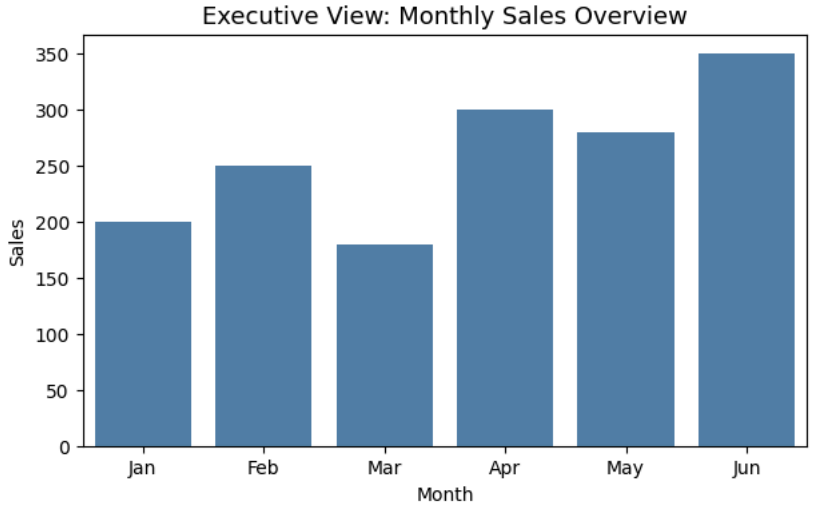
# Sample sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [200, 250, 180, 300, 280, 350],
    "Marketing_Spend": [50, 60, 45, 70, 65, 80]
}
df = pd.DataFrame(data)
df
```

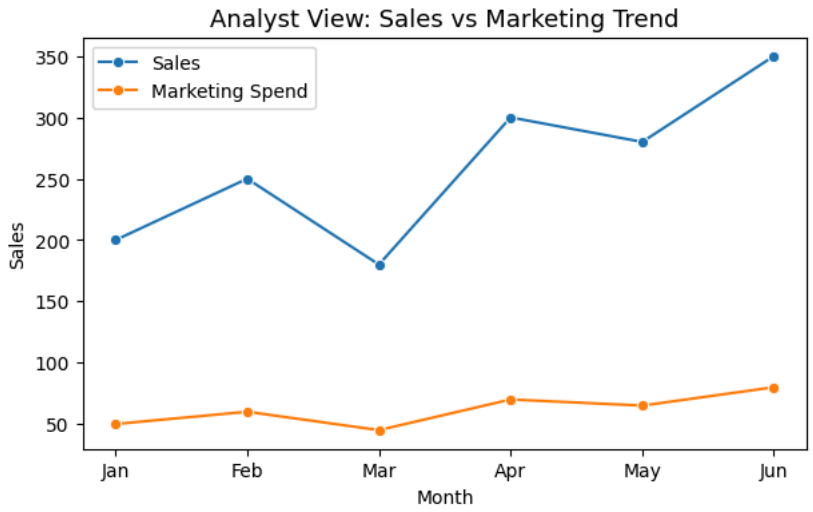
	Month	Sales	Marketing_Spend
0	Jan	200	50
1	Feb	250	60
2	Mar	180	45
3	Apr	300	70
4	May	280	65
5	Jun	350	80

Step 2. Know Your Audience — Compare Chart Styles

```
# Executive audience: prefer clarity (simple bar)
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="steelblue")
plt.title("Executive View: Monthly Sales Overview", fontsize=13)
plt.show()

# Analyst audience: prefer detailed trends (line chart)
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o", label="Marketing Spend")
plt.title("Analyst View: Sales vs Marketing Trend", fontsize=13)
plt.legend()
plt.show()
```

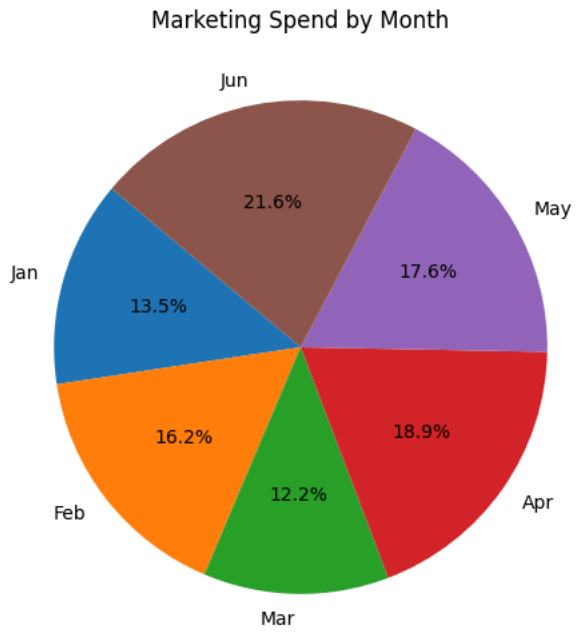




Task 1: Create one additional visualization suited for a different audience (e.g., a customer, investor, or teammate).

```
plt.figure(figsize=(6, 6))
plt.pie(
    df["Marketing_Spend"],
    labels=df["Month"],
    autopct='%1.1f%%',
    startangle=140
)

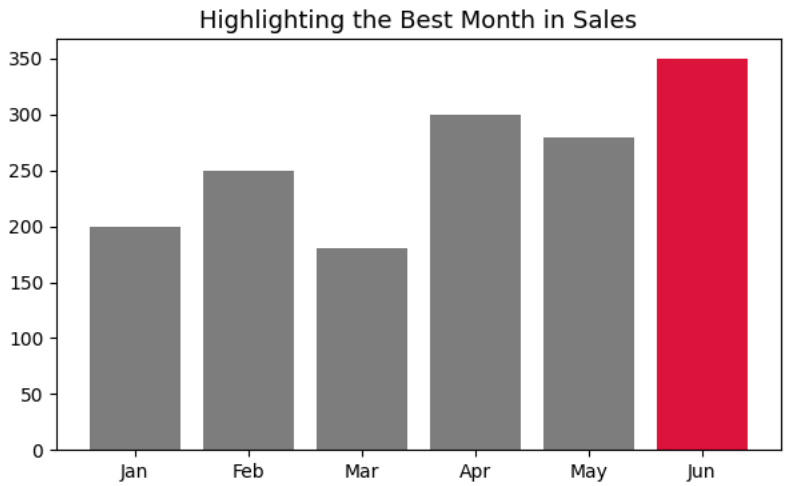
plt.title("Marketing Spend by Month")
plt.show()
```



Step 3. Know Your Message — Highlight What Matters

```
# Highlight the highest sales month (preattentive color cue)
highlight = df["Sales"].idxmax()

colors = ["gray" if i != highlight else "crimson" for i in range(len(df))]
plt.figure(figsize=(7,4))
plt.bar(df["Month"], df["Sales"], color=colors)
plt.title("Highlighting the Best Month in Sales", fontsize=13)
plt.show()
```



Task 2: Use color or annotations to emphasize a key insight (e.g., lowest point, sudden increase).

```
lowest_idx =  df["Sales"].idxmin()

df["MoM_Change"] = df["Sales"].diff()
jump_idx = df["MoM_Change"].idxmax()
largest_jump = df.loc[jump_idx, "MoM_Change"]

colors = []
for i in range(len(df)):
  if i == lowest_idx:
    colors.append ("#E74C3C")
  elif i == jump_idx:
    colors.append ("#2ECC71")
```

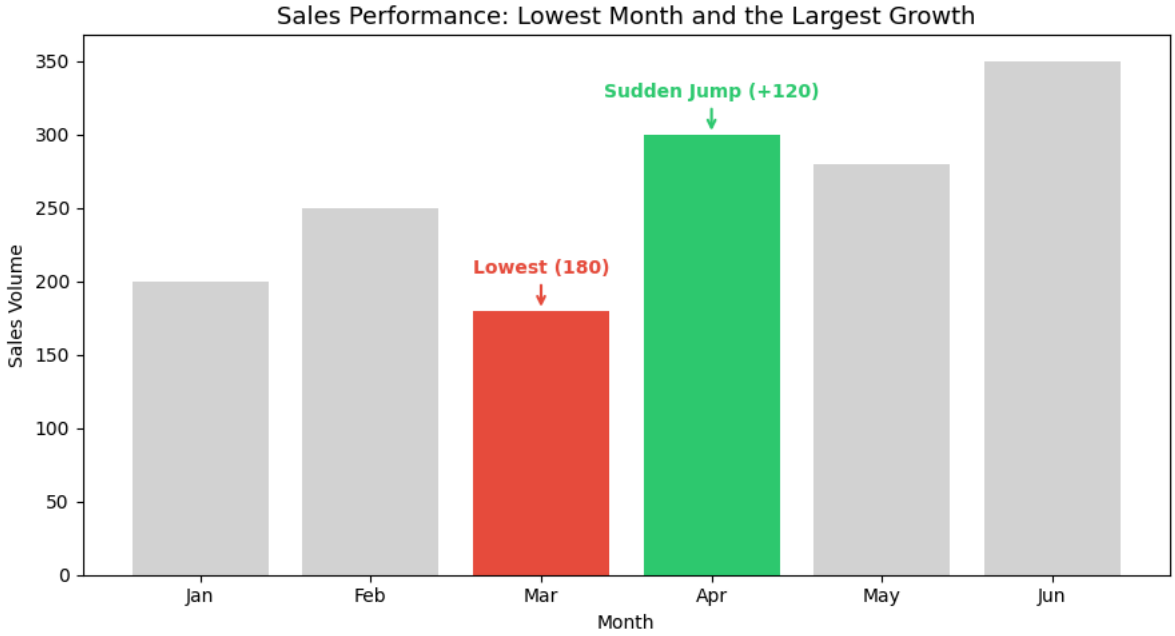
```
  else:
    colors.append ("#D3D3D3")

plt.figure(figsize=(9, 5))
plt.bar(df["Month"], df["Sales"], color=colors)

plt.annotate(
    f"Lowest ({df.loc[lowest_idx, 'Sales']})",
    xy=(lowest_idx, df.loc[lowest_idx, "Sales"]),
    xytext=(lowest_idx, df.loc[lowest_idx, "Sales"] + 25),
    ha="center",
    arrowprops=dict(arrowstyle="->", color="#E74C3C", lw=1.5),
    fontsize=10,
    fontweight="bold",
    color="#E74C3C",
)

plt.annotate(
    f"Sudden Jump (+{int(largest_jump)})",
    xy=(jump_idx, df.loc[jump_idx, "Sales"]),
    xytext=(jump_idx, df.loc[jump_idx, "Sales"] + 25),
    ha="center",
    arrowprops=dict(arrowstyle="->", color="#2ECC71", lw=1.5),
    fontsize=10,
    fontweight="bold",
    color="#2ECC71",
)

plt.title("Sales Performance: Lowest Month and the Largest Growth", fontsize=13)
plt.ylabel("Sales Volume")
plt.xlabel("Month")
plt.tight_layout()
plt.show()
```



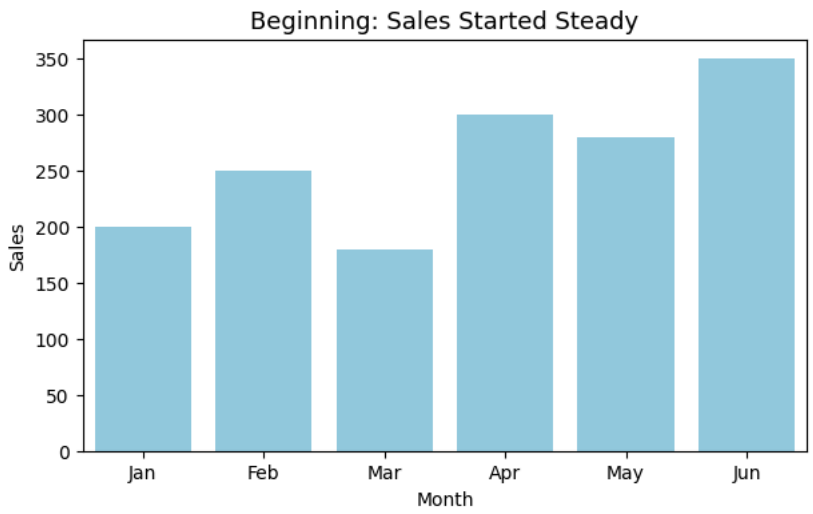
Step 4. Storytelling with Charts — Beginning, Middle, End

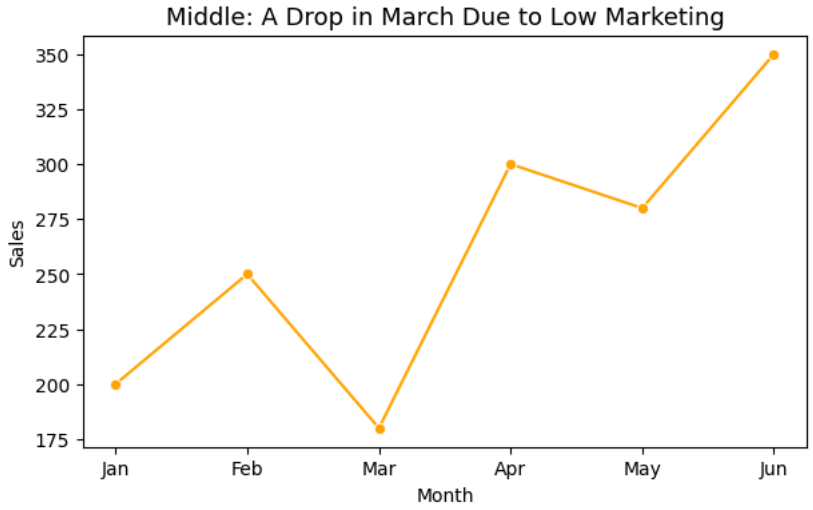
```
# Beginning: Context
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="skyblue")
plt.title("Beginning: Sales Started Steady", fontsize=13)
plt.show()

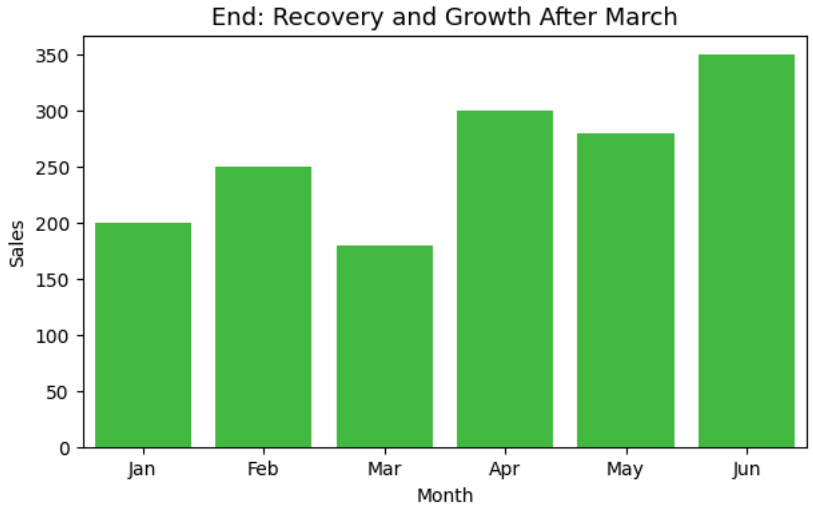
# Middle: Change or Challenge
plt.figure(figsize=(7,4))
sns.lineplot(x="Month", y="Sales", data=df, color="orange", marker="o")
plt.title("Middle: A Drop in March Due to Low Marketing", fontsize=13)
plt.show()

# End: Resolution
```

```
plt.figure(figsize=(7,4))
sns.barplot(x="Month", y="Sales", data=df, color="limegreen")
plt.title("End: Recovery and Growth After March", fontsize=13)
plt.show()
```







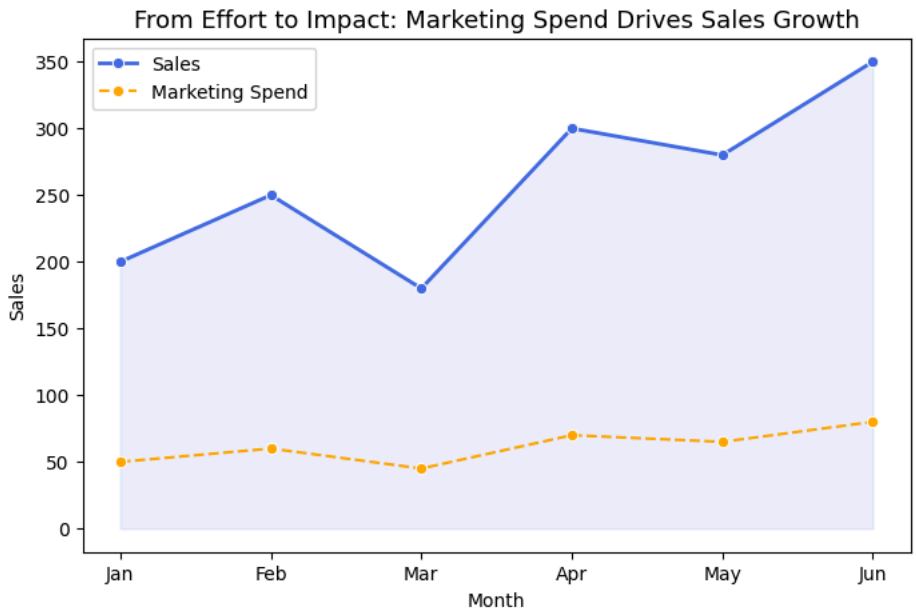
Task 3: Write one or two lines describing your data story from beginning → middle → end.

There is a Energy Drink Company that has a marketing department which spend normal for the marketing advertisements in the first two months; however, they spend a low cost in marketing advertisements during March which drops in that period but there is a recovery and growth after March because they need to spend more on marketing advertisements to gain more sales on their product and to solve their quest which is preventing their drop on the sales after the March incident.

Step 5. Bringing It All Together — Presentation Design

```
# Simple combined story chart
plt.figure(figsize=(8,5))
sns.lineplot(x="Month", y="Sales", data=df, marker="o", linewidth=2, color="royalblue", label="Sales")
sns.lineplot(x="Month", y="Marketing_Spend", data=df, marker="o", linestyle="--", color="orange", label="Marketing Spend")

plt.fill_between(df["Month"], df["Sales"], alpha=0.1, color="royalblue")
plt.title("From Effort to Impact: Marketing Spend Drives Sales Growth", fontsize=13)
plt.legend()
plt.show()
```



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5. Supplementary Activity:

1. Use a small dataset available online (kaggle, google dataset, etc.).
2. Visualize it using a bar chart.
3. Highlight one important data point (e.g., the highest or lowest value) using a distinct color.
4. Add a title and annotation that tell the story behind the data.

CODE and DATASET

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

dataset = "https://raw.githubusercontent.com/Pasioncpe/CPE-031-CPE21S1---Visualizations-and-Data-Analysis---PASION/refs/head

df = pd.read_csv(dataset)

top_5_tanks = (
    df[df["Tags"].str.contains("Tank", case=False, na=False)]
    .sort_values(by=["Base HP", "HP per lvl"], ascending=[False, False])
    .head(5)
    .reset_index(drop=True)
)

# Select required columns
result = top_5_tanks[["Name", "Tags", "Base HP", "HP per lvl"]]

# Print formatted table
print(result.to_string(index=False))
```

```
 Name         Tags  Base HP  HP per lvl
Garen Fighter,Tank      690          98
```

```
Alistar Tank,Support      685         120
    Zac Tank,Fighter      685         109
  Amumu Tank,Support      685          94
 Rammus         Tank      675          94
```

```
max_hp = top_5_tanks["Base HP"].max()
colors = ["crimson" if hp == max_hp else "lightgray" for hp in top_5_tanks["Base HP"]]

plt.figure(figsize=(8, 5))
bars = plt.bar(top_5_tanks["Name"], top_5_tanks["Base HP"], color=colors)

plt.bar_label(bars, padding=4, fontsize=11)

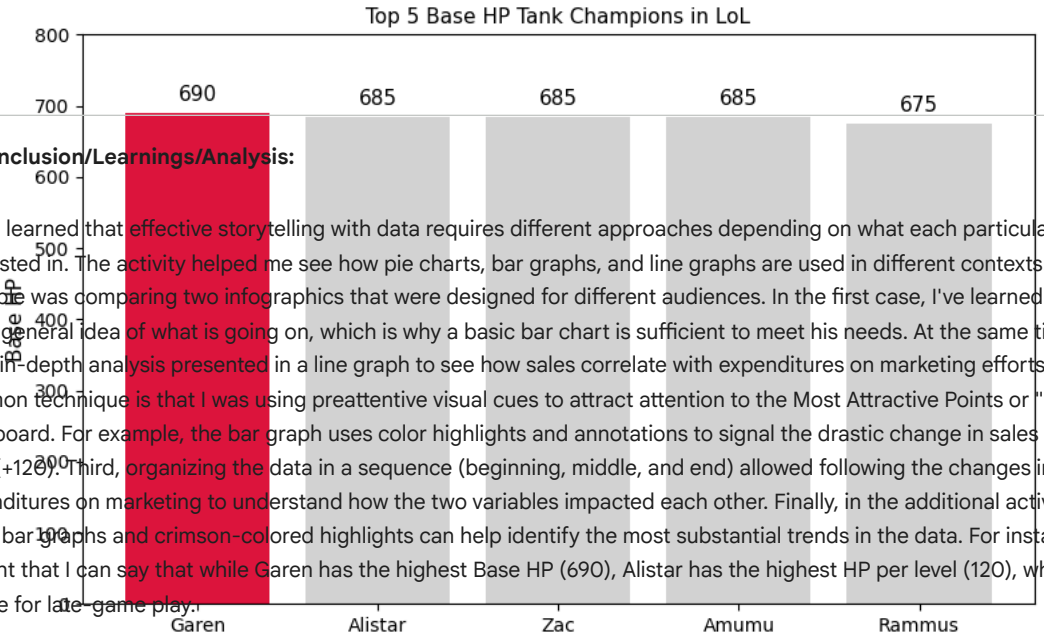
plt.ylim(0, 800)

plt.ylabel("Base HP", fontsize=11)
plt.title("Top 5 Base HP Tank Champions in LoL", fontsize=11)
plt.tight_layout()
plt.show()

max_hp_lvl = top_5_tanks["HP per lvl"].max()
colors = ["crimson" if val == max_hp_lvl else "lightgray" for val in top_5_tanks["HP per lvl"]]

plt.figure(figsize=(7, 4.5))
bars = plt.bar(top_5_tanks["Name"], top_5_tanks["HP per lvl"], color=colors)
plt.bar_label(bars, padding=3)
plt.ylim(0, max_hp_lvl * 1.15)

plt.ylabel("HP per Level")
plt.title("HP per Level for the Top 5 Base HP Tank Champions", fontsize=13)
plt.tight_layout()
plt.show()
```



6. Conclusion/Learnings/Analysis:

I have learned that effective storytelling with data requires different approaches depending on what each particular audience member is interested in. The activity helped me see how pie charts, bar graphs, and line graphs are used in different contexts. First, one useful example was comparing two infographics that were designed for different audiences. In the first case, I've learned an executive needs to see a general idea of what is going on, which is why a basic bar chart is sufficient to meet his needs. At the same time, an analyst requires more in-depth analysis presented in a line graph to see how sales correlate with expenditures on marketing efforts. Second, another common technique is that I was using preattentive visual cues to attract attention to the Most Attractive Points or "Plot Twist" from the storyboard. For example, the bar graph uses color highlights and annotations to signal the drastic change in sales from March (180) to April (+120). Third, organizing the data in a sequence (beginning, middle, and end) allowed following the changes in sales and expenditures on marketing to understand how the two variables impacted each other. Finally, in the additional activity, I have learned that using bar graphs and crimson-colored highlights can help identify the most substantial trends in the data. For instance, it becomes evident that I can say that while Garen has the highest Base HP (690), Alistar has the highest HP per level (120), which makes him the best choice for late-game play.

